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DTE Energy

10 CFR 50.73

October 9, 2017
NRC-17-0064

U. S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, D.C. 20555-0001

Reference: Fermi 2
NRC Docket No. 50-341
NRC License No. NPF-43

Subject: Licensee Event Report (LER) No. 2017-004

Pursuant to 10 CFR 50.73(a)(2)(i)(B) and 10 CFR 50.73(a)(2)(v)(B), DTE Electric Company (DTE) is submitting LER No. 2017-004, Inadequate Procedural Guidance for Residual Heat Removal Complex Ventilation Systems Leads to Condition Prohibited by Technical Specifications and Loss of Safety Function.

No new commitments are being made in this LER.

Should you have any questions or require additional information, please contact Mr. Scott A. Maglio, Manager – Nuclear Licensing, at (734) 586-5076.

Sincerely,

Keith J. Polson
Senior Vice President and CNO

Enclosure: Licensee Event Report No. 2017-004

cc: NRC Project Manager
NRC Resident Office
Reactor Projects Chief, Branch 5, Region III
Regional Administrator, Region III
Michigan Public Service Commission
Regulated Energy Division (kindschl@michigan.gov)

Enclosure to
NRC-17-0064

Fermi 2 NRC Docket No. 50-341
Operating License No. NPF-43

Licensee Event Report (LER) No. 2017-004
Inadequate Procedural Guidance for Residual Heat Removal Complex Ventilation Systems Leads to Condition Prohibited by Technical Specifications and Loss of Safety Function

NRC FORM 366 (04-2017) U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104 EXPIRES: 03/31/2020
LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)	Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

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4. TITLE

Inadequate Procedural Guidance for Residual Heat Removal Complex Ventilation Systems Leads to Condition Prohibited by Technical Specifications and Loss of Safety Function

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER
									N/A	05000
08	10	2017	2017	004	00	10	09	2017	FACILITY NAME N/A	DOCKET NUMBER 05000

9. OPERATING MODE	11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: *(Check all that apply)*			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. POWER LEVEL	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☑ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
100	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☑ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ OTHER	Specify in Abstract below or in NRC Form 366A

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT	TELEPHONE NUMBER *(Include Area Code)*
Fermi 2 / Scott A. Maglio – Manager, Nuclear Licensing	(734) 586-5076

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

14. SUPPLEMENTAL REPORT EXPECTED	15. EXPECTED SUBMISSION DATE	MONTH	DAY	YEAR
☐ YES *(If yes, complete 15. EXPECTED SUBMISSION DATE)* ☑ NO				

ABSTRACT *(Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)*

On August 10, 2017, it was determined that inadequate procedural guidance for determining operability for ventilation support systems was being utilized. The Residual Heat Removal (RHR) switchgear and pump rooms have ventilation systems to maintain operability of the equipment in the rooms. Fermi 2 procedures had directed personnel to declare the supported equipment in the rooms inoperable due to nonfunctionality of the ventilation systems only if the room temperature exceeded the operability limit. Following discovery, a review of the RHR switchgear and pump room ventilation systems for the past three years was performed. The review identified multiple instances where the ventilation systems were nonfunctional and should have resulted in entry into applicable Technical Specifications (TS). Many of these instances resulted in operations or conditions prohibited by TS, since TS Required Actions were not completed within the Completion Times for restoration of affected equipment and plant shutdown. In addition, one instance was identified where the plant configuration was such that it could have prevented the fulfillment of the safety function to remove residual heat following a design basis accident. An engineering evaluation of this specific instance was performed and verified that the plant remained within its analyzed design basis. All other instances maintained one fully operable division of heat removal equipment such that no loss of safety function existed. There were no radiological releases associated with this event. The safety significance was determined to be very low. The cause of the event was inadequate procedural guidance. Immediate actions were taken to provide interim guidance related to the procedure. Corrective actions to revise the affected procedure have been completed.

NARRATIVE

INITIAL PLANT CONDITIONS

Mode – 1
Reactor Power – 100 percent

There were no structures, systems, or components (SSCs) that were inoperable at the start of this event that contributed to this event.

DESCRIPTION OF THE EVENT

At approximately 1200 EDT on August 10, 2017, while operating in Mode 1 at 100 percent power, the DTE Electric Company (DTE) procedural guidance pertaining to the Residual Heat Removal (RHR) [BO] switchgear [SWGR] room ventilation system was questioned by the NRC Senior Resident Inspector. Technical Specification (TS) 3.8.7 requires Division I and Division II alternating current (AC) and direct current (DC) electrical power distribution subsystems [EB and EJ] to be operable in Modes 1, 2, and 3. The AC distribution system in TS 3.8.7 consists of two divisions with four 4160 V buses [BU] and four 480 V buses each. The Division I 4160 V buses 11EA and 12EB, the Division I 480 V buses 72EA and 72EB, the Division II 4160 V buses 13EC and 14ED, and the Division II 480 V buses 72EC and 72ED are all located in the RHR complex. Specifically, each bus is located in the RHR switchgear room associated with its corresponding Emergency Diesel Generator (EDG) [DG]. As described in the Updated Final Safety Analysis Report (UFSAR) Section 9.4.7.2, the four RHR switchgear rooms are each equipped with a ventilation system to dissipate the heat produced by the switchgear room equipment (e.g., the 4160 V and 480 V buses) and limit the inside ambient temperature to 104 degrees Fahrenheit (F) under all plant operating conditions. Each switchgear room ventilation system consists of an intake air duct [DUCT], high efficiency filter [FLT], and two 50 percent-capacity fans [FAN] in parallel. Thus the switchgear room ventilation systems are design features and their functionality impacts the operability of the 4160 V and 480 V buses and therefore the AC distribution subsystems. Contrary to this, a DTE procedure directed personnel to declare the 4160 V and 480 V buses inoperable per TS 3.8.7 due to nonfunctionality of the RHR switchgear room ventilation system only if the room temperature exceeded 104 degrees F. At the time of discovery of this procedural inadequacy on August 10, 2017, all the RHR switchgear room ventilation systems were functional and the associated 4160 V and 480 V buses were operable.

With a correct understanding of the applicable TS, a past operability review was performed for the three years prior to the date of discovery. This review identified 50 occurrences where an RHR switchgear room ventilation system was not functional due to maintenance activities. For each occurrence, the associated 4160 V and 480 V buses were considered operable at the time and no entry to TS 3.8.7 was made based on the unavailability of the RHR switchgear room ventilation system. Limiting Condition for Operation (LCO) 3.8.7 Condition A Required Action A.1 requires restoration of the AC electrical power distribution subsystems (i.e. the buses) to operable status within 8 hours. LCO 3.8.7 Condition C Required Action C.1 requires the reactor to be in Mode 3 in 12 hours if the Required Action and associated Completion Time of Condition A is not met. Of the 50 total occurrences described above, 25 exceeded the combined restoration Completion Time and the shutdown Completion Time (i.e. 20 hours) and therefore represent instances of an "operation or condition which was prohibited by the plant's Technical Specifications" reportable under 10 CFR 50.73(a)(2)(i)(B). Note that there is no corresponding reporting requirement under 10 CFR 50.72 and therefore no telephone notification to the NRC was made. The longest occurrence was a Division 1 RHR switchgear room ventilation system associated with EDG 11 out of service for approximately 158 hours.

The 4160 V and 480 V buses described above directly provide power for the safety-related RHR service water (RHRSW) [BI] pumps [P], Mechanical Draft Cooling Tower (MDCT) [CTW] fans [FAN], and Emergency Equipment Service Water

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(EESW) [BI] pumps, as well as other plant equipment. The TS requirements for the RHRSW pumps are found in TS 3.7.1 and the TS requirements for the MDCT fans and EESW pumps are found in TS 3.7.2. Since the 4160 V and 480 V buses were not originally declared inoperable, this supported equipment was also not declared inoperable and no entry to TS 3.7.1 and 3.7.2, or other applicable TS, was made. The Required Actions in TS 3.7.1 and 3.7.2 related to the equipment that would be affected by a single 4160 V or 480 V bus inoperable due to its ventilation system have Completion Times for restoration of the equipment that are 72 hours or longer, plus the shutdown completion time of 12 hours or longer. Therefore, any "operation or condition which was prohibited by the plant's Technical Specifications" related to exceeding the combined restoration and shutdown times of this supported equipment is bounded by those already identified as reportable under 10 CFR 50.73(a)(2)(i)(B) for the 4160 V and 480 V buses themselves.

As part of the extent of condition, it was determined that the inadequate procedural guidance was similarly applied to the RHR complex pump room ventilation systems. As described in UFSAR Section 9.4.7.3, the two RHR complex pump rooms are each equipped with a ventilation system to dissipate the heat produced by the pumps and limit the inside ambient temperature to 104 degrees F under all plant operating conditions. Each pump room ventilation system consists of an intake air duct, high efficiency filter, and two 50 percent-capacity fans in parallel. Thus the pump room ventilation systems are design features and their functionality impacts the operability of the following pumps which are located in the rooms: RHRSW pumps (two per room), EESW pumps (one per room), and EDG service water (EDGSW) pumps (two per room). At the time of discovery of the procedural inadequacy on August 10, 2017, all the RHR complex pump room ventilation systems were functional and the associated pumps were operable.

A past operability review for the pump rooms was also performed for the three years prior to the date of discovery. This review identified 17 occurrences where an RHR complex pump room ventilation system was not functional due to maintenance. For each occurrence, the associated RHRSW, EESW, and EDGSW pumps were considered operable at the time and no entry to TS 3.7.1, TS 3.7.2, or TS 3.7.8, respectively, was made based on the unavailability of the RHR complex pump room ventilation system. LCO 3.7.1 Condition C Required Action C.1 (Condition A is not applicable since both RHRSW pumps in the room are impacted) requires restoration of the RHRSW subsystem within 7 days or else shutdown to Mode 3 is required within 12 hours, respectively, per Condition D. LCO 3.7.2 Condition C Required Action C.1 requires restoration of the EESW subsystem within 72 hours or else shutdown to Modes 3 and 4 is required within 12 and 36 hours, respectively, per Condition D. LCO 3.7.8 Condition A Required Action A.1 requires declaring the associated EDGs inoperable immediately and thus entry to LCO 3.8.1. Since the EDGSW pumps were not originally declared inoperable, the supported equipment (i.e. EDGs) was also not declared inoperable and no entry to TS 3.8.1 was made. LCO 3.8.1 Condition B Required Action B.4 (Condition B is applicable since both EDGSW pumps in the room are impacted, and thus both EDGs are impacted) requires restoration of one EDG within 72 hours or else shutdown is required within 12 hours per Condition G. LCO 3.8.1 Condition A for one EDG inoperable would also apply, but its restoration time (Required Action A.6) of 14 days was never exceeded. Of the 17 total occurrences described above, two exceeded the most limiting combined restoration Completion Time and the shutdown Completion Time (i.e. 84 hours for both LCO 3.7.2 Condition D and LCO 3.8.1 Condition G) and therefore represent instances of an "operation or condition which was prohibited by the plant's Technical Specifications" reportable under 10 CFR 50.73(a)(2)(i)(B). Note that there is no corresponding reporting requirement under 10 CFR 50.72 and therefore no telephone notification to the NRC was made. The longest occurrence was a Division 2 RHR complex pump room ventilation system out of service for approximately 178 hours.

Note that TS LCO 3.8.1 Conditions A and B contains Required Actions not directly related to the EDG restoration that have Completion Times that are less limiting than the EDG restoration Completion Times. LCO 3.8.1 Required Actions A.1 and B.1 require that surveillance requirement (SR) 3.8.1.1 be performed for operable offsite power circuit(s) within one hour and once per 8 hours thereafter. SR 3.8.1.1 verifies correct breaker alignment and indicated power availability for

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each offsite circuit. Although SR 3.8.1.1 was not performed at the required time, offsite power was available during every occurrence described above and therefore the SR would have been met. LCO 3.8.1 Required Actions A.2 and B.2 state that required feature(s), supported by the inoperable EDGs, must be declared inoperable within 4 hours when the redundant required feature(s) are inoperable. LCO 3.8.1 also has a 24 hour Completion Time to either determine operable EDGs are not inoperable due to common cause failure (Required Actions A.4.1 and B.3.1) or perform SR 3.8.1.2 for operable EDGs (Required Actions A.4.2 and B.3.2). In these occurrences, the EDGs were inoperable due to the inoperability of the EDGSW pumps rather than a common cause failure of the EDGs themselves. This means that although neither Required Actions A.4.1 and B.3.1 nor Required Actions A.4.2 and B.3.2 were performed at the time they were applicable, Required Actions A.4.1 and B.3.1 would have been met in each case. LCO Condition A also has a Required Action A.3 to verify the status of Combustion Turbine Generator (CTG) [GEN] 11-1 once per 8 hours, with Required Action A.5 to restore CTG 11-1 within 72 hours from discovery of Condition A concurrent with CTG 11-1 not available. Although verification of CTG 11-1 was not performed at the required time, CTG 11-1 was available during all pump room occurrences and during all but four switchgear room occurrences, none of which exceeded the 72 hour Completion Time of Required Action A.5.

In addition, a review of all the switchgear and pump room occurrences for potential losses of safety function was performed. As described in the UFSAR Section 3.11.4.4, the RHR complex is composed of two identical divisions with the safety-related equipment in one division 100 percent redundant to that in the other division. It further states that because a separate ventilation system is provided for the switchgear rooms and pump rooms, the loss of a ventilation system does not affect safe shutdown of the plant. Therefore, no loss of safety function would exist associated with the above occurrences if the opposite division was available. All of the 67 occurrences described above were reviewed and it was determined that for each occurrence, with one exception described in detail below, the opposite division was fully available and operable such that no loss of safety function existed and the plant was fully within the UFSAR design basis.

The exception is that the RHR switchgear room ventilation system associated with Division 1 EDG 12 being out of service from May 18, 2017 at 0400 EDT to May 19, 2017 at 1744 EDT, overlapped with a degraded condition in the Division 2 RHR service water (RHRSW) subsystem. As described in LER 2017-003, submitted by DTE letter NRC-17-0048 dated July 21, 2017, the RHRSW flow control valve [FCV] E1150F068B and the Division 2 RHRSW subsystem were inoperable from May 3 to 24, 2017. The overlap period of approximately 38 hours represents an "event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to ... [r]emove residual heat" reportable under 10 CFR 50.73(a)(2)(v)(B). Note that the loss of safety function did not exist at the time of discovery such that no reporting requirement under 10 CFR 50.72 was met and therefore no telephone notification to the NRC was made. This concurrent impact on both RHRSW subsystems would require entry to LCO 3.7.1 Condition E which has a Completion Time of 8 hours. Although the overlap period exceeded the combined restoration Completion Time and the shutdown Completion Time for TS 3.7.1 (i.e. 20 hours), this occurrence was already identified above as an "operation or condition which was prohibited by the plant's Technical Specifications" reportable under 10 CFR 50.73(a)(2)(i)(B) for exceeding the combined restoration Completion Time and the shutdown Completion Time for TS 3.8.7 (i.e. also 20 hours).

SIGNIFICANT SAFETY CONSEQUENCES AND IMPLICATIONS

As discussed previously, the loss of an RHR switchgear or pump room ventilation system does not affect safe shutdown of the plant because separate ventilation systems are provided for each of the different switchgear and pump rooms. For the occurrences described above, with one exception described below, the opposite division was fully available and operable such that no loss of safety function existed and the plant was fully within the UFSAR design basis.

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An engineering evaluation was performed for the approximately 38 hours in May 2017 where the RHR switchgear room ventilation system associated with Division 1 EDG 12 being out of service coincided with a degraded condition in the Division 2 RHRSW subsystem. The evaluation determined that the available equipment at the time was adequate to fulfill applicable design functions during design basis accidents as required by the Fermi 2 licensing basis. Therefore, although this specific occurrence was considered a loss of safety function for removal of residual heat due to equipment being considered inoperable, the capability to remove residual heat was maintained throughout this period.

The engineering evaluation also documents that the RHR switchgear and pump room ventilation systems are not required for the first 24 hours of a design basis accident. This means that the pumps and 4160 V and 480 V buses that were considered inoperable as described above would have been capable of performing their intended functions for the first 24 hours even with the ventilation systems out of service. In approximately half of the occurrences described above, the out of service duration was less than 24 hours such that no loss of capability of the supported equipment (i.e. buses and pumps) would have occurred prior to restoration of the ventilation system. In addition, the engineering evaluation provides justification for alternate configurations of the RHR switchgear and pump room ventilation systems that support operability of room equipment provided the outdoor temperature is within a certain range. Examples of alternate configurations include manually blocked dampers and operation with a single fan.

UFSAR Table 3.11-4 indicates that the maximum temperature in the switchgear and pump rooms to support operability of equipment is 104 degrees F. A review of room temperatures for the periods associated with the above occurrences demonstrated that room temperatures did not exceed 104 degrees F during the times that the ventilation systems were out of service.

The Probabilistic Safety Assessment (PSA) does not model the RHR switchgear or pump room ventilation systems as the supporting analysis concluded that loss of cooling in those rooms will not affect availability of essential equipment in the rooms during the PSA mission time. As a result, risk impact of the ventilation systems being unavailable was not assessed for any of the instances described above. Risk assessment of the condition associated with the Division 2 RHRSW flow control valve (LER 2017-003) was previously performed and the condition was determined to be of very low safety significance. No change in this determination would be expected when considering the overlap of a Division 1 switchgear room ventilation system out of service with the degraded condition in the Division 2 RHRSW subsystem since the PSA model does not credit the room ventilation system.

Based on the discussion above, the safety significance of this event is very low and the event did not pose a threat to the health and safety of the public or plant personnel. There were no radiological releases associated with this event.

CAUSE OF THE EVENT

The cause of the event was inadequate procedural guidance for determining TS operability. The Fermi 2 system operating procedure for the RHR Complex Heating and Ventilation contained specific guidance on how to treat nonfunctionality of the RHR switchgear and pump room ventilation systems. This guidance incorrectly directed Operations personnel to declare the supported equipment inoperable only if the room temperature exceeded 104 degrees F. This guidance had been incorrectly implemented prior to the three year period of review documented in this LER.

The individual occurrences of the ventilation systems being out of service discussed above were related to maintenance work on the ventilation systems.

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CORRECTIVE ACTIONS

As described above, the inadequate guidance was embedded in DTE procedures. Immediate action was taken to direct Operations personnel to not utilize the procedural note regarding operability based on room temperature but rather to immediately take the TS required actions when the RHR switchgear or pump room ventilation systems were not functional. This action was an interim action pending further detailed review of the basis for the procedure. Corrective actions have since been completed to revise the affected Fermi 2 system operating procedure for the RHR Complex Heating and Ventilation. Related surveillance procedures have also been identified as requiring similar updates and the procedure revisions will be performed through the corrective action program.

The corrective actions associated with the individual occurrences of ventilation systems being out of service were to restore the ventilation systems to service following completion of maintenance.

PREVIOUS OCCURRENCES

The previous occurrences of the inadequate procedural guidance related to the RHR complex ventilation systems within the past three years were discussed in this LER. This LER also discussed overlap of this condition with that described in LER 2017-003, submitted by DTE letter NRC-17-0048 dated July 21, 2017. Since the inadequacy of the procedural guidance on RHR complex ventilation systems was not known at that time, it could not have been included in LER 2017-003.